

- p. 17, l. 3, “ $\tau = T_D$ ” should be “ $\tau = J_D$ ”
- p. 26, l. -9, in the definition of $\text{rge } S$, “ $x \in \mathbb{R}^n$ ” should be “ $x \in \mathbb{R}^m$ ”
- p. 27, l. 5, in item (b): “ $x_i \rightarrow x$ ” should be “ $x_i \rightarrow \bar{x}$ ”
- p. 29, l. 11, in Example 2:14, “ $U : \mathbb{R}^n \rightarrow \mathbb{R}^m$ ” should be “ $U : \mathbb{R}^n \rightrightarrows \mathbb{R}^m$ ”
- p. 30, l. 1, “(2.6)” should be “(2.5)”
- p. 30, l. -14, in Example 2.17 the displayed inequality should use “ K_U ” and not “ K_u ”
- p. 31, l. -7, in Exercise 2.20, “ $x_0 + u_1 + u_2$ ” should be “ $x_0 + u_0 + u_1$ ”
- p. 31, l. -1, “ $\liminf_{i \rightarrow \infty} f(x_i) \leq f(\bar{x})$ ” should be “ $\liminf_{i \rightarrow \infty} f(x_i) \geq f(\bar{x})$ ”
- p. 33, l. 11, “ $p(x_{i_k}, u_k) < m(x_{i_k}) + 1/i$ ” should be “ $p(x_{i_k}, u_k) < m(x_{i_k}) + 1/k$ ” p. 34, l. -2, “ $w \in T(S(x))$ ” should be “ $y \in T(S(x))$ ”
- p. 35, l. -14, the displayed “ $G = T \circ G \circ T$ ” should be “ $G_\rho = T \circ G \circ T$ ”
- p. 40, l. 8, “(3.2)” should be “(3.5)”
- p. 43, l. -1, “Theorem 3.9 (a)” should be “Theorem 3.9 (b)”
- p. 46, 2 lines under (3.14), “ J ” and “ J_- ” should be “ I ” and “ I_- ”
- p. 47, l. 18, “ $M(\xi)$ ” should be “ $W(\xi)$ ”, twice
- p. 47, l. -3, “Proposition 2.19” should be “Proposition 2.23”
- p. 50, l. 3, item (ii), “ $A \in \mathbb{R}^{m \times n}$ ” should be “ $A \in \mathbb{R}^{n \times m}$ ”
- p. 50, l. 3, item (ii), “ $C \in \mathbb{R}^m$ and $D \in \mathbb{R}^n$ ” should be “ $C \subset \mathbb{R}^m$ and $D \subset \mathbb{R}^n$ ”
- p. 50, l. 8, “ $x = \lambda u, y = \lambda v$ ” should “ $x = \alpha u, y = \alpha v$ ”
- p. 51, in the displayed equalities in the proof of theorem 4.4, the third entry “ $\sum_{k=0}^l (\lambda'_k + \tau \mu_k x_k)$ ” should be “ $\sum_{k=0}^l (\lambda'_k + \tau \mu_k) x_k$ ”
- p. 52, l. 5, “ $\text{con} C$ ” should be “ $\overline{\text{con} C}$ ”
- p. 54, l. 12, “ $B = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ ” should be “ $B = (0, 1)$ ”
- p. 56, l. -3, “Let be set-valued mappings and $x \in \mathbb{R}^m$.” should be “Let $x \in \mathbb{R}^m$.”
- p. 58, the second part of the proof of Proposition 4.20 has inconsistent subscripts and lacks upper limits in sums. The second part of the proof should be:

To show the opposite inclusion, fix $x \in O$ and take any $y \in F_K(x)$. Then, for every $\delta > 0$, there exist $y_\delta \in \text{con} f((x + \delta \mathbb{B}) \cap O)$ arbitrarily close to y . Note that because O is open, for all small enough δ , $x + \delta \mathbb{B} \subset O$. Considering $\delta = i^{-1}$ and large enough i , there exist $y_i \in \text{con} f(x + i^{-1} \mathbb{B})$ convergent to y . Each such $y_i = \sum_{k=0}^n \lambda_{i,k} y_{i,k}$ for some $\lambda_{i,k} \in [0, 1]$, $\sum_{k=0}^n \lambda_{i,k} = 1$, and $y_{i,k} \in f(x + i^{-1} \mathbb{B})$. Because f is locally bounded, subject to passing to a subsequence, the sequences $\lambda_{i,k}$ and $y_{i,k}$ converge as $i \rightarrow \infty$, to limits λ_k and y_k , and $\lambda_k \in [0, 1]$, $\sum_{k=0}^n \lambda_k = 1$, and $y_k \in \overline{f(x + \delta \mathbb{B})}$ for every $\delta > 0$, and so $y_k \in \bigcap_{\delta > 0} \overline{f(x + \delta \mathbb{B})}$. Also, $y = \sum_{k=0}^n \lambda_k y_k$, so that $y \in \text{con} \bigcap_{\delta > 0} \overline{f(x + \delta \mathbb{B})}$. Because the sets $(x + \delta \mathbb{B}) \cap O$ are decreasing with δ , and for small enough δ they equal $x + \delta \mathbb{B}$, $y \in \text{con} \bigcap_{\delta > 0} \overline{f((x + \delta \mathbb{B}) \cap O)}$.

- p. 61, l. 7, the display should be

$$F(x) = \left\{ (\cos \theta, x \sin \theta) : \frac{1}{x} \leq \theta \leq \frac{1}{x} + 2\pi - |x| \right\}$$

and 3 lines under the display, “ellipse become” should be “ellipse becomes”

- p. 61, Exercise 4.27 should say “Show that if $X' \subset X$ is closed and F satisfies the assumptions in Fact 4.26, the so does F' .”

- p. 64, l. -4, “ $u' = (u_1, u'_2, \dots, u'_n)$ ” should be “ $u' = (u'_1, u'_2, \dots, u'_n)$ ”
- p. 72, l. 1, “ $\phi_k(t) := x_i + (t - t_i)v_0$ ” should be “ $\phi_k(t) := x_i + (t - t_i)v_i$ ”
- p. 75, l. 3, “ $t \in [0, \bar{\tau} - \varepsilon]$ ” should be “ $t \in [0, \bar{\tau} + \varepsilon]$ ”
- p. 91, l. 18, “ $w(x) = A$ ” should be “ $w(x) = 0$ ”
- p. 91, l. -16, in Exercise 6.13, “ $\{x \in O \mid V(x) \leq r\}$ ” should be “ $\{x \in O \mid w(x) \leq r\}$ ”
- p. 94, l. 24, in Exercise 6.19, “conclusions in (c), (e)” should be “conclusions in (e)”
- p. 98, the first display after Exercise 7.1 should be

$$\sqrt{(\gamma x_1 + \lambda y_1)^2 + (\gamma x_2 + \lambda y_2)^2} \leq \gamma \sqrt{x_1^2 + x_2^2} + \lambda \sqrt{y_1^2 + y_2^2}.$$

The second one should be

$$x_1 y_2 + x_2 y_2 \leq \sqrt{x_1^2 + x_2^2} \sqrt{y_1^2 + y_2^2},$$

- p. 102, l. 16, in the next to last line of Example 7.8, “ $T_{[1,2] \times [3,\infty)}((2,5)) = (-\infty, \infty) \times (-\infty, \infty)$ ” should be “ $T_{[1,2] \times [3,\infty)}((2,5)) = (-\infty, 0] \times (-\infty, \infty)$ ”
- p. 102, l. -5, “Exercise 7.10 or 7.9” should be “Exercise 7.8 or Exercise 7.9”
- p. 103, l. -13, in the next to last line of the proof of Proposition 7.11, “which contradicts $w \in (T_C(x))^-$ and $v \in T_C(x)$ ” should be “which contradicts $w \in (T_C(x))^- = N_C(x)$ and $v \in (N_C(x))^-$ ”
- p. 105, l. -5, “Exercise 7.13.” should be “Exercise 7.13:”
- p. 106, l. 13, “such that $z + (v, 0) \in \text{epi } f$ for all $t \geq 0$ ” should be “such that $z + t(v, 0) \in \text{epi } f$ for all $t \geq 0$ ”
- p. 108, l. 1, “ $\nabla f(x) \subset \partial f(x)$. In fact, $\nabla f(x) = \partial f(x)$ ” should be “ $\nabla f(x) \in \partial f(x)$. In fact, $\{\nabla f(x)\} = \partial f(x)$ ”
- p. 108, l. -13, “and thus by (b)” should be “and thus by (c)”
- p. 110, l. 1, “ $\partial f(0) = \text{con}\{(1,0), (-1,0), (0,1), (0,-1)\} = [-1,1] \times [-1,1]$ ” should be “ $\partial f(0) = \text{con}\{(1,0), (-1,0), (0,1), (0,-1)\}$ ”
- p. 113, l. 19, the last line before Proposition 7.33, “can be show” should be “can be shown”
- p. 114, l. -7, the last line of the proof of Corollary 7.34, “Exercise 6.17” should be “Exercise 6.15”
- p. 120, l. 8, “any convergent sequence $(x_i, y_i) \in M$ ” should be “any convergent sequence $(x_i, y_i) \in \text{gph } M$ ”
- p. 120, l. -13, the three-line displayed calculation is nonsense, and it should be:

$$\begin{aligned} (x - \bar{x}) \cdot (y - \bar{y}) &= (1 - \lambda)(x - \bar{x}) \cdot (y - y_1) + \lambda(x - \bar{x}) \cdot (y - y_2) \\ &\geq 0. \end{aligned}$$

- p. 121, Proposition 8.9 (b), “ $\dot{\phi}$ is nonincreasing” should be “ $\|\dot{\phi}\|$ is nonincreasing”
- p. 122, l. 3 and l. 4, the inequalities should be in the opposite direction.
- p. 124, l. -6, “by (7.6) and Exercise 8.22” should be “by (7.6)”
- p. 125, l. 2, in (8.20), it should be “ $\dot{x} = F_k(x) := -\nabla f(x) - k(x - P_C(x))$ ”. Two lines further, in l. 4, it should say “Since $-F_k$ is the gradient of a convex function”.
- p. 125, l. 15, in the 4th line of the displayed formula for $2\alpha_k(t)\dot{\alpha}_k(t)$, “ $-2(\phi_k(t) - P_C(\phi_k(t)))\nabla f(\phi_k(t))$ ” should be “ $-2(\phi_k(t) - P_C(\phi_k(t))) \cdot \nabla f(\phi_k(t))$ ”
- p. 129, l. -7, in the display, “ $\gamma(t)$ ” should be “ $\gamma_k(t)$ ”
- p. 134, l. 13, “ $\lambda \in [0, 1]$ ” should be “ $\alpha \in [0, 1]$ ”, or all occurrences of α in the proof should be λ
- p. 135, in the proof of Theorem 9.2, there are 4 related changes: l. 6, “ $s := \sup \{y \cdot x \mid y \in \partial f(x)\}$ ” should be “ $s(v) := \sup \{y \cdot x \mid y \in \partial f(x)\}$ ”; l. 7, the displayed “ $f'(x; \bar{v}) > s$ ” should be “ $f'(x; \bar{v}) > s(\bar{v})$ ”; l. 9, in (9.3) the second inequality should be “ $\bar{y} \cdot \bar{v} > s(\bar{v})$ ”; and on l. 15, it should say “Consequently, $f'(x, v) = s(v)$.”
- p. 137, in the proof of Theorem 9.4, there are 3 related changes: l. 13, the display should be “ $f^\circ(x; v) \geq m(v) := \max \{y \cdot v \mid y \in \partial^\circ f(x)\}$ ”; l. 14 should have “ $f^\circ(x; \bar{v}) > m$ ”; l. 15, the display’s second inequality should be “ $\bar{y} \cdot \bar{v} > m(\bar{v})$ ”; and l. 18 should end with “ $f^\circ(x; v) = m(v)$.”

- p. 136, l. 8, “which leads to $g''(0; v) = v$ for $v > 0$. Similar arguments shows $g''(0; v) = -v$ for $v < 0$.” should be “which leads to $g^\circ(0; v) = v$ for $v > 0$. Similar arguments shows $g^\circ(0; v) = -v$ for $v < 0$.”
- p. 139, l. 10, “at almost $t \in [0, T]$ ” should be “at almost all $t \in [0, T]$ ”
- p. 141, l. 20, “ $\text{con}\{(1, -1), (1, 1)\} = [-1, 1] \times \{1\}$ ” should be “ $\text{con}\{(1, -1), (1, 1)\} = \{1\} \times [-1, 1]$ ”
- p. 147, l. 16, under (10.1), “processes” should be “process”
- p. 148, l. 16, under Exercise 10.1, “mappings” should be “mapping”
- p. 152, l. 2, “is *robust* if there exists a continuous function $\rho : B(A) \rightarrow [0, \infty)$ ” should be “is *robust* if the basin of attraction $B(A)$ is a neighborhood of A and there exists a continuous function $\rho : B(A) \rightarrow [0, \infty)$ ”
- p. 152, l. 17, at the end of Example 10.7, “contains $-x_2^2$ and x_2^2 ” should be “contains $(-x_2^2, 0)$ and $(x_2^2, 0)$ ”
- p. 157, l. 3, “has $\|Ae\| \leq \delta'$ and” should be “has”, i.e., the comment on $\|Ae\|$ is unnecessary
- p. 157, l. 4 and l. 7, the displays should end with “ $\|x\|/16$ ” and not “ $\|x\|/8$ ”, and consequently the first, the second, the third, and the fifth “ $\|x\|/8$ ” in the display in l.11-15 should be “ $\|x\|/16$ ”
- p. 160, l. 8, “ $\frac{d}{dt}\nabla f(\phi(t)) = -\|\nabla f(\phi(t))\|^2$ ” should be “ $\frac{d}{dt}f(\phi(t)) = -\|\nabla f(\phi(t))\|^2$ ”
- p. 160, l. -7, “mode” should be “model”
- p. 169, l. -7 and l. -5, “ λ ” should be “ α ”
- p. 184, l. 4, “ $((t_{j-1}, j], j)$ ” should be “ $((t_{j-1}, t_j], j)$ ”
- p. 186, l. -20, in Exercise 12.8, “Example 2.2” should be “Example 12.4”
- p. 187, l. -12, “for all $I > i_\epsilon$ ” should be “for all $i > i_\epsilon$ ”
- p. 188, proof of Theorem 12.8 uses should not use K as a subscript in j_{K^i} because K is a compact set in the theorem. Once one replaces all K in subscripts with M , in line -2, “ $\phi_i(\cdot, j_K^i)$ ” should be “ $\phi_i(\cdot, j_{M^i})$ ”
- p. 190, l. 5, “and $H(x) = \emptyset$ otherwise” should be “and $H(x) = \{x\}$ otherwise”
- p. 192, l. 16, “Any such solution is also a solution to (12.3)” should be “Any such solution is also a solution to (12.10)”

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